

Phase 0 Computational Setup, Simulation Progress, and Modeling Strategy

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1 Phase 0 Objective

Phase 0 is designed to compare representative baseline tubular architectures under controlled and identical simulation conditions. The broader screening matrix includes an organic or biomimetic tubular structure, a hybrid organic–inorganic tubular structure, and a predominantly inorganic tubular structure. The objective is to identify which baseline architecture maximizes physical plausibility, computational tractability, structural stability, confined-water retention, and measurable response under external field perturbation.

The initial validated simulation branch focuses on a predominantly inorganic BN-like tubular scaffold with confined water. This branch is used as the first reference architecture because it provides a mechanically robust, geometrically well-defined tubular domain that can be directly modified to test wall charge, wall polarity, neutralization, and field-response controls.

2 Computational Environment

The initial computational setup is developed on macOS. This environment is used for documentation, geometry generation, small test systems, scripting, post-processing, plotting, and visualization. Larger production simulations may later be migrated to a Linux workstation, HPC system, or cloud compute environment depending on system size, number of replicas, trajectory length, and computational cost.

3 Software Stack and Justification

3.1 LAMMPS

LAMMPS is available locally through the executable `lmp_mpi`. The installed version is LAMMPS 10 Dec 2025, compiled with Open MPI v5.0.7. The build includes several packages relevant to this project, including MOLECULE, KSPACE, DIPOLE, DRUDE, DIELECTRIC, MANYBODY, REAXFF, QEQ, RIGID, SPIN, PHONON, BROWNIAN, COLLOID, CORESHELL, and DPD.

LAMMPS is selected as the primary engine for inorganic, hybrid organic–inorganic, coarse-grained, and field-driven models because it provides broad support for particle-based materials modeling, long-range electrostatics, dipolar interactions, external fields, rigid bodies, many-body potentials, reactive force fields, and specialized packages for dielectric, spin, and phonon-related models. Its flexibility is particularly relevant for Phase 0 because the candidate architectures include inorganic tubes, hybrid coated tubes, organic or biomimetic tubes, confined polar media, and externally driven response tests.

The current macOS LAMMPS build is suitable for local setup, geometry validation, debugging, and small-to-medium test systems. Because OpenMP is not enabled at the compiler level and no GPU acceleration is reported, larger production simulations may be more efficiently executed later on a Linux workstation or HPC/cloud environment.

3.2 GROMACS

GROMACS is available locally through the executable `gmx`. The installed version is GROMACS 2025.4, compiled in mixed precision with thread-MPI, OpenMP support, AVX2 SIMD instructions, and FFTW. GPU support is disabled in the local build.

GROMACS is selected for biomolecular, peptide-like, organic, and soft-matter candidates because it is highly optimized for classical molecular dynamics of water-containing molecular systems and provides mature tools for energy minimization, equilibration, trajectory generation, and standard structural analyses. It is particularly suitable if the organic or biomimetic Phase 0 candidate is represented as a peptide-like, aromatic peptide-like, or other biomolecular tubular assembly.

The current macOS GROMACS build is suitable for local testing and moderate CPU-based simulations. Large-scale production runs may benefit from optimized Linux or GPU builds.

3.3 Python Analysis Environment

Python 3.12.12 is available through the Miniforge environment. Python is used for geometry generation, workflow automation, trajectory analysis, dipole-orientation metrics, autocorrelation functions, plotting, tabular summaries, and preparation of report-ready figures. The initial package list includes NumPy, SciPy, pandas, matplotlib, MDAnalysis, MDTraj, Jupyter, and tqdm.

3.4 Later-Stage Tools

Quantum ESPRESSO and Octopus are reserved for later electronic-structure and optical-response modules, including DFT, TDDFT, transition-dipole calculations, polarizability estimates, and fragment-level electronic parameterization. COMSOL is reserved for later continuum electromagnetic and device-level modeling once effective dielectric, polarizability, or conductivity parameters are available from molecular simulations, electronic calculations, or literature data.

4 Repository Structure

The repository is organized to separate documentation, references, scripts, systems, controls, simulations, results, figures, logs, and daily notes. Simulation inputs are stored under `simulations/`, structural data and generated trajectories under `systems/`, processed numerical outputs under `results/`, and report-ready plots under `figures/`. Daily progress is tracked in `notes/`. Heavy output files such as `.data`, `.lammprtrj`, and visualization files are intended to be tracked through Git LFS.

5 Candidate Architecture Definition

The first validated geometric reference is a tubular scaffold with outer diameter 24.0 nm, lumen diameter 14.0 nm, wall thickness 5.0 nm, and segment length 20.0 nm. This geometry is used as a common reference for Phase 0 so that the organic or biomimetic, hybrid organic–inorganic, and predominantly inorganic candidates can be compared under identical dimensional constraints.

The initial generated scaffold is a geometric point-cloud representation that was then used to construct force-field-ready simulation branches. Geometry inspection confirmed an estimated lumen diameter of 14.0 nm, outer diameter of 24.0 nm, segment length of 20.0 nm, and 17,280 shell points.

Table 1: Initial Phase 0 candidate matrix.

Class	Initial candidate	Primary purpose
Organic or biomimetic	Peptide-like or aromatic peptide tubular scaffold with polar wall groups	Test biomimetic confinement, hydration-layer behavior, polar wall effects, and structural persistence.
Hybrid organic-inorganic	BNNT or CNT scaffold with polar or peptide-like coating	Test whether a mechanically robust inorganic tube plus polar coating improves confined-water ordering and dielectric response.
Predominantly inorganic	BN-like tubular scaffold with confined water	Test robust tubular confinement, confined-water ordering, dipole autocorrelation, and field-induced response.

6 Control Systems

The Phase 0 comparison requires controls to determine whether the measured response is caused by structured confinement, confined water, wall chemistry, wall charge, or ordered functionalization. The initial controls are:

- dry or empty scaffold, to isolate the role of confined water;
- water-filled BN-like polar scaffold, to establish the first inorganic reference;
- water-only contained reference, to separate scaffold effects from finite-domain confinement;
- carbon-like neutral scaffold-water control, to test a neutral matched tubular scaffold;
- BN-neutralized scaffold-water control, to isolate the effect of BN-like wall charge while preserving BN-like geometry and B/N atom-type topology;
- field-free and axial-field versions of the water-containing controls.

7 Simulation Protocol

The Phase 0 simulation protocol was implemented as a staged molecular simulation workflow using LAMMPS. The first production-ready baseline was the predominantly inorganic BN-like tubular scaffold with confined water. The scaffold geometry was kept fixed during the initial water-equilibration and screening simulations so that differences among systems could be attributed primarily to scaffold chemistry, scaffold charge patterning, and field response rather than to scaffold deformation.

All directly compared water-containing systems used the same nominal geometry, box dimensions, water count, and contained finite-domain protocol. The common simulation box was defined over the following Cartesian bounds:

$$x, y \in [-140, 140] \text{ \AA}, \quad z \in [-120, 120] \text{ \AA}.$$

The nominal lumen radius used for analysis was 70 Å, the nominal outer scaffold radius was 120 Å, and the nominal analyzed axial half-length was 100 Å.

The principal systems analyzed in the Day 005 charge and polarity control matrix were:

- BN-like polar scaffold-water reference;
- BN-neutralized scaffold-water control;
- carbon-like neutral scaffold-water control;
- water-only contained reference.

The BN-like polar scaffold-water reference contains 17,280 scaffold atoms and 30,000 water molecules, for a total of 107,280 atoms. The water-only reference contains the same 30,000 water molecules but no scaffold. The BN-neutralized control preserves the BN-like scaffold geometry and B/N atom-type topology, but sets the scaffold charges to zero. The carbon-like neutral control preserves the same scaffold geometry and water configuration, but converts the scaffold to a neutral carbon-like placeholder wall.

All contained molecular dynamics tests used a timestep of 0.10 fs, SHAKE-constrained water, reflective finite-domain walls, and a water thermostat at 300 K. Field-free matched-time controls were analyzed at 20,000 steps. Axial electric-field diagnostics were run for 10,000 steps using:

$$E_z = 0.01 \text{ V/Å},$$

implemented in LAMMPS real units as 0.230605 kcal/(mol Å e).

8 Analysis Metrics

The Phase 0 analysis emphasized stability, radial confinement, confined-water structure, and water dipole response. The primary scalar metrics extracted from each trajectory were:

- mean radial water oxygen position, $\langle r_O \rangle$;
- maximum radial water oxygen position, $r_{O,\max}$;
- mean axial absolute position, $\langle |z_O| \rangle$;
- fraction of water oxygens inside the nominal lumen segment;
- fraction of water oxygens radially outside the nominal lumen radius;
- mean axial dipole projection, $\langle \cos(\theta_z) \rangle$;
- mean absolute axial dipole projection, $\langle |\cos(\theta_z)| \rangle$;
- axial dipolar order parameter, S_z .

The axial dipolar order parameter was computed as:

$$S_z = \frac{1}{2} \left(3\langle \cos^2(\theta_z) \rangle - 1 \right),$$

where θ_z is the angle between the water dipole vector and the axial z -direction. For an approximately isotropic orientation distribution, $\langle |\cos(\theta_z)| \rangle \approx 0.5$ and $S_z \approx 0$.

The quantitative Day 005 screening table is stored in `results/phase0/day005_charge_polarity_screening_s`

9 Results

9.1 BN-like Scaffold-Water Baseline

The BN-like polar scaffold-water baseline was successfully stabilized using a contained finite-domain protocol. The validated system contained 17,280 scaffold atoms and 30,000 confined water molecules. The 300 K contained trajectory reached 20,000 steps without lost atoms, missing bonds, NaNs, or dangerous neighbor-list builds.

In the field-free 20,000-step control, the BN-like scaffold maintained a maximum radial water oxygen position of approximately 73.34 Å. This value remained close to the nominal lumen boundary and was substantially lower than the corresponding values observed in the neutralized and carbon-like controls.

9.2 Matched Charge and Polarity Controls

Day 005 introduced two matched scaffold controls to separate geometric confinement from wall polarity and charge-pattern effects. The BN-neutralized control preserved the BN-like geometry and B/N atom-type topology but removed the scaffold charges. The carbon-like control preserved the same tubular geometry and water configuration, but replaced the scaffold chemistry with a neutral carbon-like placeholder.

The field-free 20,000-step comparison showed that the BN-like polar scaffold maintained the strongest radial confinement. In contrast, both neutral scaffold controls allowed much larger radial excursions. The BN-neutralized scaffold reached a maximum radial water oxygen position of approximately 95.25 Å, and the carbon-like neutral scaffold reached approximately 97.12 Å. The water-only contained control reached approximately 89.84 Å.

Table 2: Final field-free Day 005 metrics at 20,000 steps.

System	$r_{\text{O,max}}$ (Å)	$\langle r_{\text{O}} \rangle$ (Å)	$\langle \cos \theta_z \rangle$	S_z
BN-like polar scaffold-water	73.34	41.79	0.0024	-0.0030
BN-neutralized scaffold-water	95.25	40.72	0.0026	-0.0049
carbon-like neutral scaffold-water	97.12	40.76	0.0035	-0.0020
water-only contained	89.84	40.88	0.0044	-0.0049

The water dipole orientation remained approximately isotropic in all field-free systems. The final S_z values were close to zero, and the final mean axial dipole projections were also near zero. Therefore, the dominant architecture-dependent difference in the field-free simulations was radial confinement rather than spontaneous axial dipole ordering.

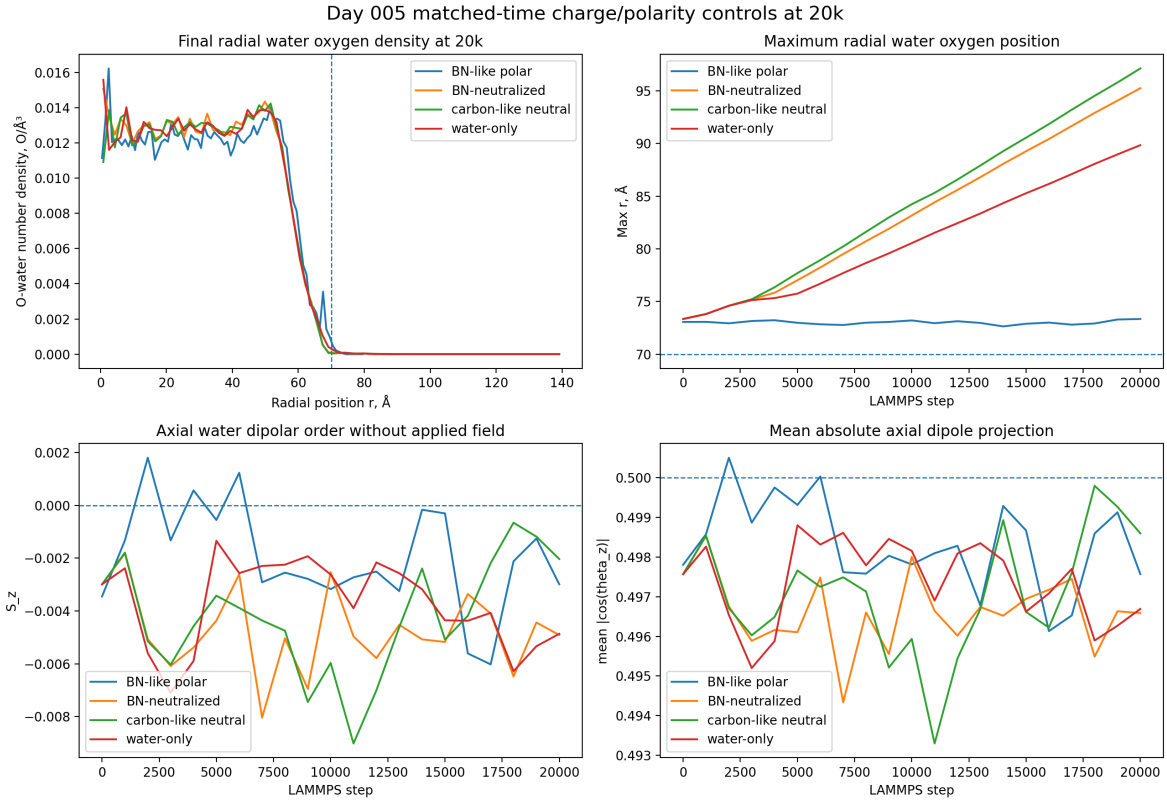


Figure 1: Matched-time field-free charge and polarity controls at 20,000 steps. The BN-like polar scaffold maintains strong radial water confinement, whereas BN-neutralized and carbon-like neutral controls allow substantially larger radial excursions.

9.3 Axial Field-Response Matrix

A short axial electric-field diagnostic was performed for all four water-containing systems using $E_z = 0.01 \text{ V/\AA}$. All systems showed a strong and similar axial water-dipole response. The final mean axial dipole projection was approximately 0.59 in all cases, and the final S_z values were approximately 0.26.

The architecture-dependent difference again appeared in radial confinement. The BN-like polar scaffold maintained the maximum radial water oxygen position near $72.82 \text{ \AA}$ under fieldZ. By contrast, the BN-neutralized scaffold reached $105.12 \text{ \AA}$, the carbon-like neutral scaffold reached $107.78 \text{ \AA}$, and the water-only control reached $97.15 \text{ \AA}$.

Table 3: Final Day 005 axial field-response metrics after 10,000 fieldZ steps.

System	$r_{O,\max} \text{ (\AA)}$	$\langle r_O \rangle \text{ (\AA)}$	$\langle \cos \theta_z \rangle$	S_z
BN-like polar scaffold-water + fieldZ	72.82	41.50	0.5880	0.2567
BN-neutralized scaffold-water + fieldZ	105.12	40.25	0.5891	0.2592
carbon-like neutral scaffold-water + fieldZ	107.78	40.26	0.5949	0.2605
water-only contained + fieldZ	97.15	40.36	0.5962	0.2652

These results indicate that the applied axial field dominates the average axial dipole alignment of water, while scaffold charge and polarity control radial confinement.

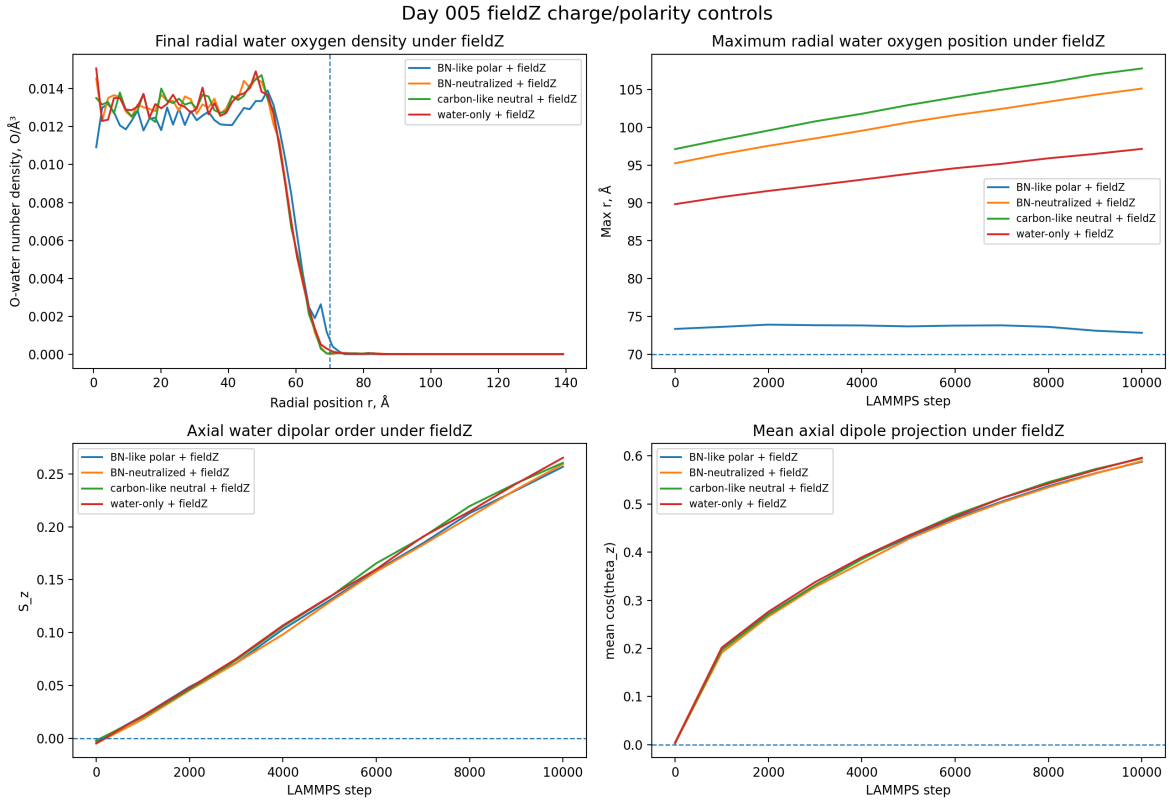


Figure 2: Axial field-response charge and polarity controls. All systems show similar field-induced axial water-dipole alignment, but only the BN-like polar scaffold preserves strong radial confinement under fieldZ.

10 Discussion

The Day 005 charge and polarity matrix provides the first direct Phase 0 evidence that the BN-like polar scaffold is not simply acting as a geometric tube. The BN-like polar scaffold, the BN-neutralized scaffold, and the carbon-like neutral scaffold share the same nominal tubular geometry and water loading. However, only the BN-like polar scaffold maintains strong radial confinement over the screened trajectories.

The BN-neutralized scaffold is the most direct control because it preserves the BN-like geometry and B/N atom-type topology while removing scaffold charge. Its loss of confinement relative to the BN-like polar scaffold indicates that wall charge and polarity are key design variables. The carbon-like neutral control supports the same conclusion because it also fails to reproduce the radial confinement behavior of the BN-like polar scaffold.

The field-response simulations show that all water-containing systems respond similarly to the applied axial field in terms of mean axial dipole projection and S_z . Therefore, on the current short diagnostic timescale, the field controls average axial dipole alignment, while the wall charge and polarity pattern controls radial confinement. This distinction is important for Phase 0 because it separates a confinement-stability criterion from a field-response criterion.

These results should be interpreted as short-timescale screening diagnostics rather than converged dielectric-response calculations. Longer trajectories, replicate runs, improved force-field validation, and weaker-field linear-response tests will be required before extracting quantitative susceptibility, relaxation time, or dielectric-response parameters.

11 Next-Step Decision Criteria

The current Phase 0 decision criterion is whether a candidate architecture can maintain stable confined water while producing measurable and interpretable dipole-response metrics. Based on the Day 005 screening matrix, the BN-like polar scaffold is the strongest current inorganic baseline because it preserves radial confinement under both field-free and fieldZ conditions.

The next decision points are:

1. extend the BN-like polar scaffold trajectories to longer timescales;
2. run replicate trajectories to separate systematic architecture effects from trajectory noise;
3. test weaker electric fields to approach a linear-response regime;
4. introduce a hybrid organic–inorganic coated scaffold while preserving the same geometric reference;
5. define the organic or biomimetic tubular candidate under the same lumen, hydration, and temperature constraints.

The immediate Phase 0 ranking criterion is not total energy alone, but the combined behavior of structural stability, radial confinement, water dipole statistics, and controlled response under external axial field.